A Brief Guide to the iSTAY R Code for Figures 1–4

This guide describes the R code used to reproduce Figures 1–4 of the Chao et al. (2025) manuscript, including the stability and synchrony profiles, biodiversity–stability relationships, and mixed-effects model analyses based on data from the Jena Experiment. The corresponding R script contains the calculation and plotting functions used to generate these figures. All data and R code are available on Zenodo and in Anne Chao's GitHub repository at https://github.com/AnneChao/MS_iStay.

Requirements and data

R version 4.1.0 or later is required. Install the packages listed below before running the script. The Jena datasets are distributed with the iSTAY package.

```
library(tidyverse)

library(lmerTest)

library(gridExtra)

library(ggpubr)

library(iSTAY)

data("Data_Jena_76_community_populations")

data("Data_Jena_20_metacommunities")
```

Data_Jena_76_community_populations contains 76 plot-level datasets used in Figures 1–3. Data_Jena_20_metacommunities contains 20 metacommunity datasets used in Figure 4. The figure script also defines the required plotting and LMM functions.

Stability and synchrony functions

The iSTAY_Single() function calculates information-based temporal stability of individual series. The iSTAY_Multiple() function calculates gamma stability, alpha stability, beta stability, and synchrony for multiple time series.

```
iSTAY_Single(data, order.q = c(1, 2), Alltime = TRUE,
             start_T = NULL, end_T = NULL)
iSTAY_Multiple(data, order.q = c(1, 2), equal_weights = TRUE,
               Alltime = TRUE, start_T = NULL, end_T = NULL)
```

Argument	Description
data	A vector, matrix, data frame, or (for iSTAY_Multiple) a list of datasets; sampling units are rows and time points are columns.
order.q	Positive orders of stability or synchrony; default c(1, 2).
equal_weights	TRUE assigns equal weights to sampling units; FALSE uses biomass weights. Applies to iSTAY_Multiple.
Alltime	TRUE uses all time points; FALSE selects a specified range.
start_T, end_T	Starting and ending time columns when Alltime = FALSE.

Figure 1. Gamma, alpha, and synchrony profiles

Figure 1 compares two Jena plots, B1A04 (four species) and B4A14 (two species). Panel (a) shows plot-level gamma stability, panel (b) shows species-level alpha stability together with individual species profiles, and panel (c) shows within-plot species synchrony. All profiles are evaluated over q = 0.1–2 with increments of 0.01. Biomass weighting is used for the multiple-series calculations.

Figure 1a–c. R code

```
df1 <- list(
  B4A14_2 = Data_Jena_76_community_populations$B4A14_B4_2,
  B1A04_4 = Data_Jena_76_community_populations$B1A04_B1_4
)
df1 <- Map(function(x, nm) {
  rownames(x) <- paste0(nm, rownames(x))
  x
}, df1, names(df1))
output_fig_1 <- iSTAY_Multiple(df1, order.q = seq(0.1, 2, 0.01),
                               equal_weights = FALSE)
output_fig_1b <- list(
  B1A04_4 = iSTAY_Single(df1$B1A04_4, order.q = seq(0.1, 2, 0.01)),
  B4A14_2 = iSTAY_Single(df1$B4A14_2, order.q = seq(0.1, 2, 0.01))
) |> bind_rows(.id = "Site")
fig_1a(output_fig_1)
fig_1b(output_fig_1, output_fig_1b)
fig_1c(output_fig_1)
```

The selected plots are stored in df1. Individual species stability profiles from iSTAY_Single() are overlaid in panel (b). The script combines the three panels into Figure_1.pdf.

Figure 2. Biodiversity–stability and biodiversity–synchrony relationships within plots

Figure 2 relates logarithmic species richness to (a) gamma stability, (b) alpha stability, and (c) within-plot species synchrony, using the 76 plot datasets. Analyses are performed for q = 0.5, 1, and 2 using biomass weights. Each linear mixed-effects model includes log2 species richness as a fixed effect and block-specific random intercepts and slopes.

Figure 2a–c. R code

```
split_names2 <- str_split(names(Data_Jena_76_community_populations),
                          "_", simplify = TRUE)
structure2 <- data.frame(
  block = split_names2[, 2],
  log2_sowndiv = log2(as.numeric(split_names2[, 3]))
)
output2 <- iSTAY_Multiple(Data_Jena_76_community_populations,
                          order.q = c(0.5, 1, 2), equal_weights = FALSE)
output_fig_2a <- LMM_2_to_4(output2, structure2, "Gamma")
output_fig_2b <- LMM_2_to_4(output2, structure2, "Alpha")
fig2_or_4(output_fig_2a, "Gamma")
fig2_or_4(output_fig_2b, "Alpha")
```

For panel (c), monoculture plots (species richness = 1, log2_sowndiv = 0) are excluded. The structural metadata and the corresponding output rows are filtered consistently for all three orders q, preserving dataset alignment.

```
keep2 <- which(structure2$log2_sowndiv != 0)
structure2c <- structure2[keep2, , drop = FALSE]
output2c <- output2[unlist(lapply(0:2, function(j)
  keep2 + j * nrow(structure2))), ]
output_fig_2c <- LMM_2_to_4(output2c, structure2c, "Synchrony")
fig2_or_4(output_fig_2c, "Synchrony")
```

The three panels are assembled in Figure_2.pdf.

Figure 3. Temporal variation across 10-year windows

Figure 3 examines temporal changes in the effects of species richness. The left panels show average gamma stability, alpha stability and species synchrony grouped by species richness and q; the right panels report the corresponding fixed-effect slopes of log2 species richness from the mixed-effects models. Equal weighting is used.

The analysis uses 12 overlapping 10-year windows: the first contains 2003 and 2005–2013; the remaining windows begin in 2005–2015. The year 2004 is excluded.

Figure 3a–c. Left panels

```
split_names3 <- str_split(names(Data_Jena_76_community_populations), "_", simplify = TRUE)
year_windows <- lapply(2003:2015, function(start) {
  yrs <- if (start == 2003) c(2003, 2005:2013) else start:(start + 9)
  if (2004 %in% yrs) return(NULL)
  as.character(yrs)}) |> purrr::compact()
names(year_windows) <- as.character(c(2003, 2005:2015))
output_fig_3_left <- lapply(names(year_windows), function(start_year) {
  biomass_data <- lapply(Data_Jena_76_community_populations,
    function(df) df[, year_windows[[start_year]], drop = FALSE])
  result <- iSTAY_Multiple(biomass_data, order.q = c(0.5, 1, 2),
                           equal_weights = TRUE)
  result |> mutate(Start_year = as.numeric(start_year),
                   log2_sowndiv = rep(log2(as.numeric(split_names3[, 3])), 3))})
Summary_fig_3_left <- bind_rows(output_fig_3_left) |>
  mutate(Order_q = paste0("q = ", Order_q)) |>
  group_by(Start_year, Order_q, log2_sowndiv) |>
  summarise(mean_gamma = mean(Gamma), mean_alpha = mean(Alpha),
            mean_syn = mean(Synchrony), .groups = "drop")
fig_3_left(Summary_fig_3_left)$Gamma_plot
fig_3_left(Summary_fig_3_left)$Alpha_plot
fig_3_left(Summary_fig_3_left)$Synchrony_plot
```

The synchrony panel excludes monocultures, while gamma and alpha stability include all richness levels.

Figure 3d–f. Right panels

```
structure3 <- data.frame(
  block = split_names3[, 2],
  log2_sowndiv = log2(as.numeric(split_names3[, 3]))
)
output_fig_3a_right <- slope_3(metric_name = "Gamma")
output_fig_3b_right <- slope_3(metric_name = "Alpha")
output_fig_3c_right <- slope_3(metric_name = "Synchrony")
fig_3_right(output_fig_3a_right)
fig_3_right(output_fig_3b_right)
fig_3_right(output_fig_3c_right)
```

The slope_3() function fits the block-level mixed-effects model separately for each window and order q. Monocultures are excluded from the synchrony slope analysis, with the output and structural metadata kept aligned. All six panels are assembled in Figure_3.pdf.

Figure 4. Biodiversity–stability and spatial synchrony among plots

Figure 4 uses the 20 Jena metacommunities to show the relationships between log2 plot richness and (a) gamma stability, (b) alpha stability, and (c) spatial synchrony among plots. Analyses use biomass weighting and q = 0.5, 1, and 2.

Figure 4a–c. R code

```
split_names4 <- str_split(names(Data_Jena_20_metacommunities),
                          "_", simplify = TRUE)
structure4 <- data.frame(
  block = split_names4[, 1],
  log2_sowndiv = log2(as.numeric(split_names4[, 2]))
)
output4 <- iSTAY_Multiple(Data_Jena_20_metacommunities,
                          order.q = c(0.5, 1, 2), equal_weights = FALSE)
output_fig_4a <- LMM_2_to_4(output4, structure4, "Gamma")
output_fig_4b <- LMM_2_to_4(output4, structure4, "Alpha")
output_fig_4c <- LMM_2_to_4(output4, structure4, "Synchrony")
fig2_or_4(output_fig_4a, "Gamma")
fig2_or_4(output_fig_4b, "Alpha")
fig2_or_4(output_fig_4c, "Synchrony")
```

Output

Run the accompanying R script from beginning to end to create the figures directory and export Figure_1.pdf, Figure_2.pdf, Figure_3.pdf, and Figure_4.pdf. The analytical definitions, fitted models, and graphical parameters are specified in the script. The R version and package versions can be recorded with sessionInfo().